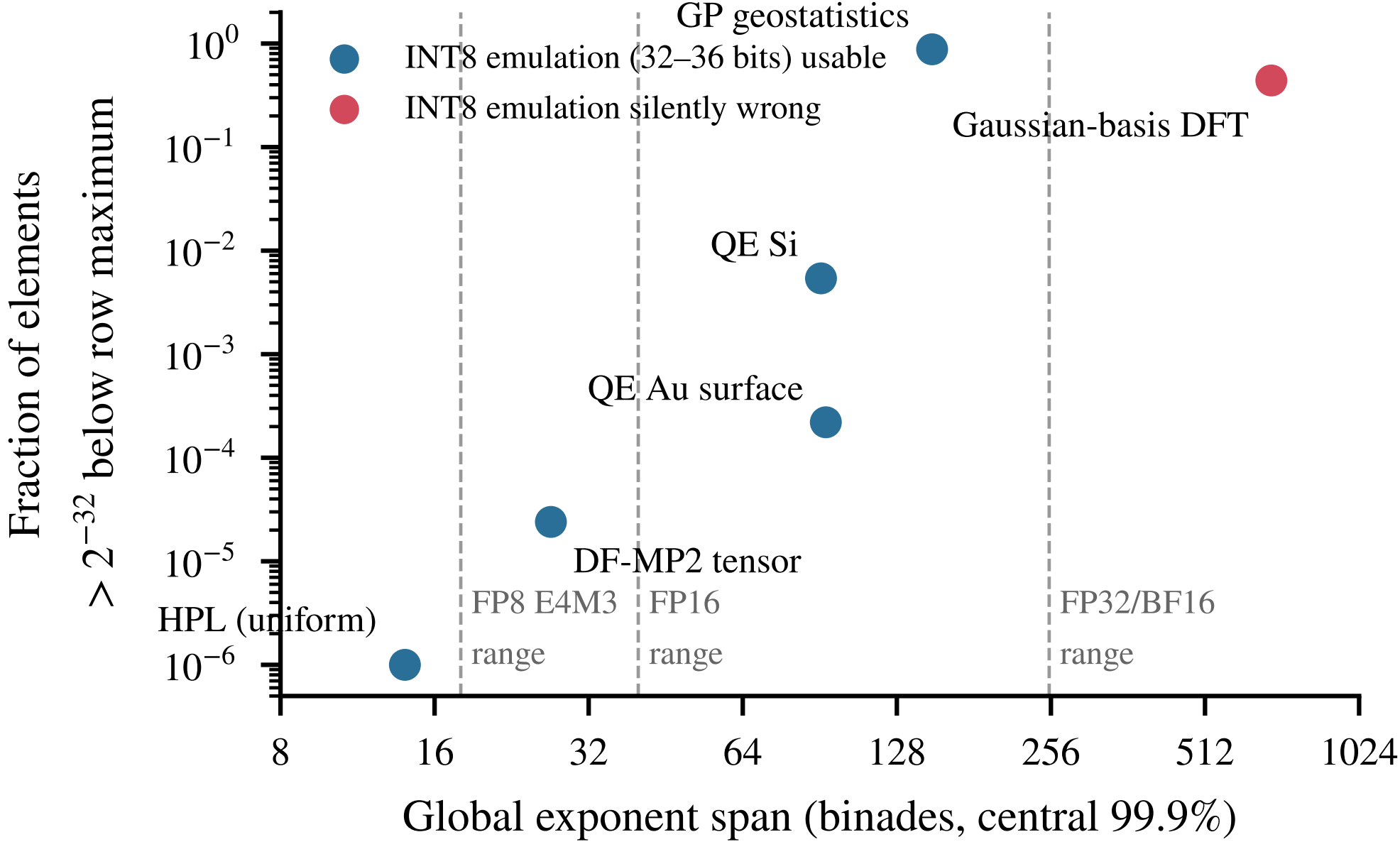
GP geostatistics
INT8 emulation (32–36 bits) usable
INT8 emulation silently wrong
Gaussian-basis DFT
QE Si
QE Au surface
DF-MP2 tensor
HPL (uniform)
FP8 E4M3 range
FP16 range
FP32/BF16 range
Fraction of elements > $2^{-32}$ below row maximum
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
$10^{-6}$
8
16
32
64
128
256
512
1024
Global exponent span (binades, central 99.9%)